Supriyo Banerjea

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.96
Methodist School Dankuni	Class XII (ISC)	2020	90.75%
Methodist School Dankuni	Class X (ICSE)	2018	93.80%

EXPERIENCE

Data Scientist Intern -- *Macromill Inc., Shinagawa, Tokyo*

- Built a **leakage-free forecasting pipeline** to predict survey response completion and automate a previously manual invitation-sizing workflow, producing forecasts in **under one second**.
- Designed a **hierarchical response-time model** combining pre-send survey features with partial response observations across six prediction windows from 0 to 48 hours.
- Implemented a **Markovian Beta-Binomial forecasting model** with calibrated cure-fraction estimation, hourly Bayesian hazard priors, **PyMC/ADVI hierarchical learning**, and JAX-accelerated MAP-Laplace inference to update forecasts as responses arrived.
- Backtested the system on **500 historical surveys** using frozen cohorts and temporally valid evaluation, achieving **93.5-99.8% empirical coverage** for 95% prediction intervals.
- Projected a **15-20% reduction in over- and under-invitation**, with potential savings in panel rewards and reduced respondent fatigue.

Research Assistant -- *Speech Information Processing Lab, IIT Hyderabad*

- Built and integrated **ROS 2-based streaming audio perception pipelines** for a DRDO-funded research project, covering sound-event detection, keyword spotting, and audio-source localization.
- Optimized deep audio models for **NVIDIA Jetson AGX Orin** through shared preprocessing, FP16 inference, ONNX/TensorRT deployment, and PyTorch graph compilation.
- Developed an **open-vocabulary template-matching pipeline** for detecting target sounds in acoustic mixtures, with emphasis on low-latency inference, memory efficiency, and edge deployment.

PROJECTS

EyeRAGent: Multimodal Self-Correcting Ophthalmic Clinical Assistant *(in collaboration with LVPEI)*

- **Built an End-to-End Multimodal RAG System:** Developed an ophthalmic clinical assistant supporting text, voice (*faster-whisper*), and image inputs using **MedGemma 1.5-4B**, backed by a curated knowledge base of **21,635 medical chunks** from textbooks, PubMed, and clinical guidelines.
- **Engineered the Ophthalmic Vision Pipeline:** Combined MedGemma-based routing across **11 ophthalmic imaging modalities** with fine-tuned **EyeCLIP** and isolated-prefix softmax for modality-specific zero-shot image classification.
- **Designed Hybrid Retrieval and Reranking:** Integrated **BM25** and **MedEmbed** dense retrieval through ChromaDB, followed by **MedCPT cross-encoder reranking** and source-aware weighting to prioritize authoritative medical literature.
- **Implemented Clinical Safety Guardrails:** Developed emergency triage, deterministic anatomy-graph validation, and claim-level grounding with self-correction, achieving **100% anatomical fidelity** and a **95.8% grounding pass rate** on the project evaluation set.
- **Developed Longitudinal Patient Memory:** Built confidence-decayed session state and persistent patient memory using **MemPalace and SQLite**, with contradiction-aware updates and clinician summaries; achieved a **4.14/5 external-judge score** with zero pipeline failures during evaluation.

H2-NAFNet: Physics-Informed Joint Super-Resolution & Denoising

- Developed a **2x joint super-resolution and denoising model** for noisy, single-channel sensor data.
- Adapted a SIDD-pretrained NAFNet architecture for grayscale inputs using **bilinear upsampling and channel replication** to preserve fine spatial details.
- Designed a **physics-informed heteroscedastic data-consistency loss** that projects high-resolution predictions back to low-resolution space to reduce noise artifacts.
- **Scaled training** with PyTorch DDP and **synthetic corruption augmentations** to improve robustness against unseen compound degradations.

- Optimized inference using **test-time augmentation** and **consistency refinement** to maximize output quality.

RESEARCH

Publication: M. Nani, **S. Banerjea**, R. Raghavan, "*Towards Robust Subject Identification From EEG Segments With Data Augmentation Techniques*." DOI

SKILLS

Programming & ML: Python, C++, SQL, PyTorch, transformers, CLIP, JAX, scikit-learn, PyMC, Hugging Face

Data & Tools: pandas, NumPy, Databricks, PySpark, Git, Linux, Weights & Biases, Streamlit

Vector DB & Search: ChromaDB, BM25, MedEmbed, Hybrid Search, Knowledge Graphs

ACHIEVEMENTS

- **Winner, ICASSP 2025 Grand Challenge (GC-5)** -- EEG-Music Emotion Recognition.
- **Runner-up, KLA AI Hackathon, IITH 2026** -- achieved Top 10 on the private leaderboard (0.883 score) with a physics-informed NAFNet image restoration model.
- **Top 15, Meltwater Hackathon 2026** -- ranked among *150+ teams from IITH and IIITH*; designed an ML model to play a Hangman-style game.

RELEVANT COURSEWORK

Deep Learning, Probability, Statistical Inference, Practical Challenges in Image Analysis, Bayesian Data Analysis, Linear Algebra, and Data Structures & Algorithms

POSITIONS OF RESPONSIBILITY

Teaching Assistant: *Pattern Recognition and Machine Learning (EE2802)*